



Software Requirements Specification

Style Sense AI - Inclusive AI Fashion Stylist with Measurement
and Color-Blind Safety Analysis

R26-ISE-001

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1.Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to describe the requirements, functionalities, system features, and overall behavior of the AI-based Intelligent Fashion Assistance System. This document acts as a guide for developers, testers, researchers, and stakeholders to understand the proposed system and its expected functionalities during development. The system aims to improve the online fashion and clothing selection experience by using Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Natural Language Processing (NLP), and image processing technologies.

The proposed solution includes four main components designed to address various fashion-related challenges users face. The first component features an AI-based virtual size recommendation system. It extracts body measurements from uploaded full-body images using pose estimation techniques and accurately recommends suitable clothing sizes. The second component offers an intelligent text-driven fashion recommendation system. It understands user preferences through natural language input and generates personalized outfit suggestions based on occasions, style, color, and clothing preferences.

The third component presents an AI-driven color assistance system tailored for users with Color Vision Deficiency (CVD). This part identifies clothing colors and patterns, improves pattern visibility, and creates CVD-specific visual simulations to help users understand how clothing appears based on their color perception issues. The fourth component emphasizes adaptive outfit matching for CVD users. It analyzes top and bottom clothing combinations, assesses color compatibility, and recommends visually appropriate outfit combinations to enhance fashion accessibility and user confidence.

The main goal of this system is to offer a smart, accessible, accurate, and user-friendly fashion assistance platform. It supports personalized recommendations, improves clothing selection decisions, reduces fitting and return issues, and increases accessibility for users with

different visual and fashion-related needs. This SRS document outlines the functional requirements, non-functional requirements, system constraints, user interactions, and performance expectations necessary for successfully implementing the proposed system.

1.2 Document Conventions

This Software Requirement Specification (SRS) document is prepared using a clear and structured format to ensure easy understanding for all stakeholders involved in the project. Headings and subheadings are used to organize the content into different sections such as functional requirements, non-functional requirements, system descriptions, and user requirements. Important terms and technical concepts related to Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, and Color Vision Deficiency are written in a consistent manner throughout the document.

The word “system” refers to the proposed AI-based Intelligent Fashion Assistance System developed in this project. Terms such as “user” refer to individuals who interact with the system to receive fashion recommendations, size predictions, color assistance, and outfit matching support. Functional requirements describe the services and features provided by the system, while non-functional requirements describe system quality attributes such as performance, usability, reliability, and security.

All requirements included in this document are written in simple and understandable language to maintain clarity and consistency. Diagrams, tables, and figures are used where necessary to improve readability and provide a better understanding of the proposed solution. English is used as the primary language throughout the document, and standard software engineering documentation practices are followed during the preparation of this SRS.

1.3. Intended Audience and Reading Suggestions

Audience	Purpose
Developers	Understand the functional requirements, system architecture, algorithms, and technical implementation details of all system components.
Project Managers	Analyze the project scope, milestones, timelines, risks, and implementation boundaries.
QA Engineers	Design test cases and verify that all functional and non-functional requirements are properly covered.
End Users	Understand the overall capabilities of the intelligent fashion assistance system, including size recommendation, color assistance, outfit matching, and personalized fashion recommendations.

1.4. Product Scope

The AI-Powered Virtual Fashion Assistant is an advanced software tool created to aid online fashion choices by integrating artificial intelligence, machine learning, computer vision, and natural language processing methods. The system aims to tackle typical issues encountered by online buyers, such as trouble choosing the right clothing size, finding appropriate outfits, comprehending clothing colors and patterns for individuals with Color Vision Deficiency (CVD), and obtaining tailored fashion suggestions in accordance with user tastes.

Online shoppers frequently face doubt when buying apparel because they cannot physically try on items, resulting in wrong size choices, uncoordinated outfits, and elevated return rates. These difficulties are much more pronounced for individuals with CVD, who may find it hard to accurately differentiate clothing colors and patterns. Current e-commerce platforms generally offer restricted accessibility features and do not include smart tools that adjust to specific user requirements and preferences.

To tackle these challenges, the suggested system offers a cohesive fashion support platform that incorporates four key AI-driven elements. The initial element gathers body dimensions from complete images utilizing YOLO-based pose estimation and suggests the most suitable clothing

size. The second element utilizes natural language processing to interpret user preferences expressed in free text and creates fashion recommendations that are aware of the context. The third element offers color identification, pattern detection, CVD-focused simulation, and image enhancement to aid users with challenges in color perception. The fourth element assesses outfit compatibility and suggests visually appropriate clothing pairs customized for various CVD types. The platform allows users to submit clothing and body pictures, articulate their fashion needs in everyday language, and obtain immediate, tailored suggestions via an intuitive interface. The solution boosts shopping confidence, enhances accessibility, lowers return rates, and fosters inclusive online fashion experiences for a variety of users by combining multiple AI techniques into one platform.

1. AI-Based Virtual Intelligent Size Recommendation

This component addresses the challenge of selecting the correct clothing size in online shopping environments. Using a custom-trained YOLO pose estimation model, the system detects 17 human body key points from a single full-body image and extracts measurements such as shoulder width, hip width, and estimated body height. Pixel-based distances are converted into approximate real-world measurements and compared with standardized clothing size charts to recommend the most suitable size. In addition, the module analyzes body proportions to provide personalized body appearance insights. This approach reduces reliance on manual measurements, fitting rooms, and size-related product returns while supporting a wide range of body types, genders, and plus-size users.

2. Intelligent Text-Driven Fashion Recommendation

This component enables users to describe their clothing preferences in natural language, such as occasion, color, style, and fit requirements. Natural Language Processing (NLP) techniques are used to interpret the free-text input and convert it into structured recommendation criteria. The system combines semantic similarity analysis, occasion awareness, and size filtering to retrieve and rank the most relevant fashion items. Unlike keyword-based recommendation systems, this module understands contextual meaning and generates personalized and explainable suggestions. As a result, users can quickly identify suitable outfits for weddings, office wear, parties, and other events while reducing the time spent searching large product catalogs.

3. AI-Driven Color Assist for Color Vision Deficiency

This component is specifically designed to support users with Color Vision Deficiency (CVD) in understanding clothing colors and patterns. A Convolutional Neural Network (CNN) identifies the dominant clothing color and predicts confidence scores for supported color classes. A second CNN model classifies pattern types such as stripes, checks, floral, and plain. The system then applies image enhancement techniques including Contrast Limited Adaptive Histogram Equalization (CLAHE) and Canny edge detection to improve pattern visibility. Users can select their CVD type—Protanopia, Deuteranopia, or Tritanopia—and receive personalized simulated outputs that reflect how the clothing appears under their specific condition. This module enhances accessibility and enables users to make more confident clothing selections independently.

4. Adaptive Outfit Matching for Color Vision Deficiency Users

This component evaluates whether selected top and bottom clothing items form a visually compatible outfit for users with different types of Color Vision Deficiency. The system detects dominant colors from both garments, analyzes color harmony using CVD-aware compatibility rules, and generates an outfit matching score. If the selected combination is not suitable, the system recommends alternative color pairings that provide better visual distinction and harmony. By considering the user's specific CVD type, this module delivers personalized outfit suggestions that improve confidence and reduce uncertainty when coordinating clothing combinations.

The proposed solution is implemented as an integrated web-based platform that combines computer vision, deep learning, image processing, and natural language processing technologies. Users can upload images, provide text-based preferences, and receive real-time intelligent recommendations through a unified and user-friendly interface. By addressing size selection, personalized recommendation, color accessibility, and outfit coordination within a single platform, the system delivers a comprehensive and inclusive fashion assistance solution for a diverse range of online shoppers.

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2. Overall Description

2.1 Product Perspective

The proposed system, called the Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis, is an intelligent fashion assistance platform powered by artificial intelligence. It helps users choose appropriate clothing by considering body dimensions, personal style preferences, intended occasions, and color vision capabilities. By integrating several AI-driven components into one cohesive system, it aims to enhance both the precision and inclusivity of online fashion recommendations.

Access begins with a secure login process, where users create an account and sign in to enter the platform. Once logged in, they land on the home page, which serves as a gateway to various AI-supported styling tools. The first key feature is the AI-Based Virtual Intelligent Size Recommendation Module. Here, users upload a full-body photo, and the system applies a specially trained YOLO pose estimation model to identify key body landmarks. From these points, it automatically calculates measurements such as shoulder width, hip width, and approximate height. Rather than relying on manual inputs, this module uses image analysis to recommend the best clothing size. It also offers tailored advice on appearance and styling based on the user's body shape and proportions. Following the size assessment, users indicate whether they have normal color vision or Color Vision Deficiency (CVD).

This choice determines the subsequent recommendation path. For users with normal color vision, the system directs them to the Intelligent Text-Driven Fashion Recommendation Module. They can describe their desired clothing using everyday language specifying style, colors, event type, or preferences. Leveraging Natural Language Processing (NLP) and semantic matching techniques, the system interprets these inputs and suggests outfit options that align with the user's size and context.

If a user identifies as having CVD, the AI-Driven Color Assist Module becomes active. It first detects the specific type of color vision deficiency such as protanopia, deuteranopia, or tritanopia. Based on this classification, the Adaptive Outfit Matching Module for CVD Users evaluates uploaded clothing items to assess whether color combinations are distinguishable and suitable for

the user's visual condition. It then offers alternative, more recognizable color pairings and provides confident, accessible outfit suggestions.

Designed as an independent AI-powered platform, the system works across web and mobile devices. It brings together computer vision, machine learning, NLP, image analysis, and recommendation algorithms into a single, unified solution. Additionally, it is built to support future integration with e-commerce sites, online apparel retailers, and smart retail systems, aiming to boost accessibility, personalization, and overall satisfaction in digital fashion shopping.

2.2 Product Functions

The proposed system offers a range of core functionalities designed to work in tandem, delivering intelligent and personalized fashion assistance for both typical users and those with Color Vision Deficiency (CVD).

The first component is user registration and authentication. Individuals can create accounts, log in securely, and manage their personal profiles. This enables the system to store user-specific data, such as size details, preferences, and past recommendations for more tailored future interactions.

Next, the system features an AI-powered body measurement and size prediction process. Users upload a full-body image, which is analyzed by a custom-trained YOLO-based pose estimation model to detect key body landmarks. Using pixel-distance calculations, the system derives measurements including shoulder width, hip width, and estimated height, then determines the best clothing size. The process accommodates various genders, body types, and plus sizes, enhancing inclusivity and recommendation precision. A related feature provides personalized body appearance suggestions based on the analysis.

Users also select their category either standard or color-blind during setup. This choice triggers distinct recommendation and analysis pathways tailored to their needs. For standard users, the system includes a text-based outfit recommendation engine. Users describe their needs using natural language, such as event type, preferred colors, styles, or overall aesthetic. An NLP module interprets the input's meaning and matches it against fashion datasets using semantic similarity techniques, returning customized outfit suggestions aligned with both user preferences and predicted size.

Additionally, the system incorporates occasion-aware recommendations. Outfits are ranked according to specific contexts like weddings, office settings, casual outings, festivals, parties, or formal events, increasing the relevance and usability of the suggestions. For users with CVD, the system identifies the specific type of color vision deficiency and saves this information to inform future personalization efforts.

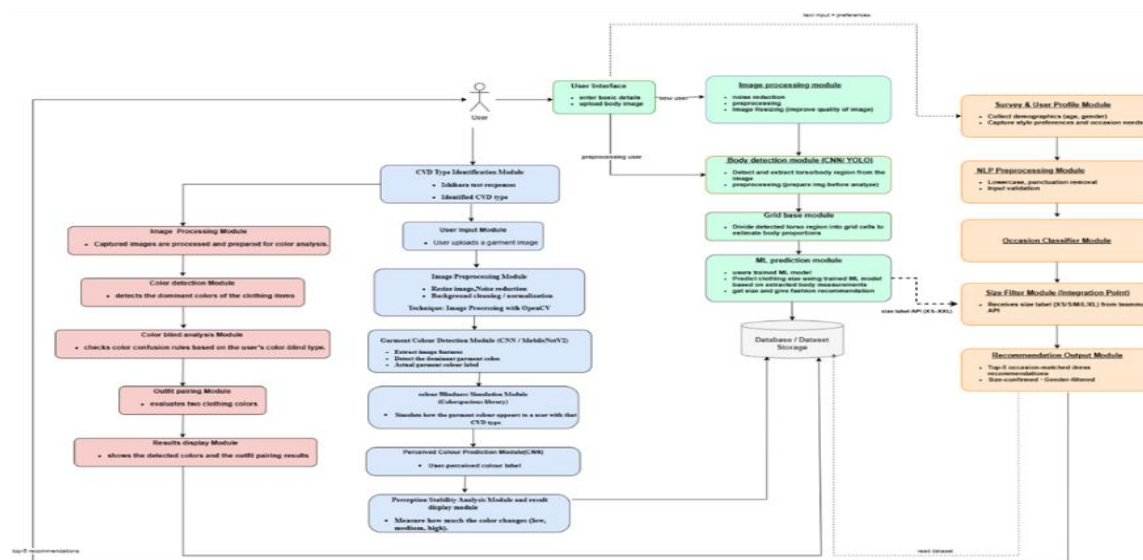
It also includes an AI-based clothing color and pattern analysis tool tailored for color-blind users. When clothing images are uploaded, the system uses image processing to detect dominant colors and patterns, enhances pattern contrast, and simulates how colors appear under various CVD conditions.

The system identifies potential issues related to color-blindness and, when problematic color combinations are found, suggests alternative outfit pairings that offer improved visibility and contrast. It features a real-time response capability, delivering quick analysis and recommendations to enhance usability and overall user experience.

Moreover, it allows users to submit feedback and rate suggestions, supporting ongoing improvements in recommendation accuracy and personalization through continuous learning and refinement of the underlying dataset.

A key function for these users is adaptive outfit matching, where the system assesses whether chosen clothing combinations are visually distinguishable and aesthetically appropriate based on the user's color vision profile.

This is a simplified high-level architecture diagram for our proposed system.



2.3 User Classes and Characteristics

The system serves a diverse range of users, each with distinct needs, technical abilities, and accessibility considerations.

The main group consists of everyday fashion consumers who have typical color vision. Their primary use of the system involves receiving reliable size suggestions, tailored outfit recommendations, and style advice suited to personal preferences and specific occasions. Since many in this group are not technically proficient, the interface must be intuitive, straightforward, and visually clear. These users value fast results, precise sizing estimates, and seamless support when choosing outfits during online shopping.

A key secondary group includes individuals with Color Vision Deficiency (CVD), who often struggle to differentiate certain colors and feel uncertain when coordinating clothing. This group may include people with protanopia, deuteranopia, or tritanopia. To address their needs, the system offers features such as color-blind-friendly outfit pairing, enhanced color contrast, pattern clarity improvements, and customized visual feedback. The design for these users prioritizes accessibility, clarity, and the ability to make clothing choices independently.

The system also caters to fashion-savvy individuals and frequent online shoppers seeking smart styling suggestions for events like work functions, weddings, parties, festivals, or casual outings. These users look for recommendations that reflect current trends and suit their individual tastes and context-specific requirements.

Another category comprises researchers, developers, and system administrators tasked with maintaining, refining, and assessing the platform. These users handle data curation, track AI model accuracy, refine recommendation logic, and ensure system stability and security. They generally have expertise in machine learning, software engineering, and data systems.

In the future, the system may also serve e-commerce platforms and fashion retailers aiming to embed the recommendation engine into their digital storefronts. These users will require scalable APIs, effective personalization tools, and capabilities that enhance customer interaction through intelligent fashion support.

Overall, the system is built to serve users across varying levels of technical skill, fashion awareness, accessibility needs, and shopping behaviors, while ensuring usability, inclusiveness, and personalized experiences.

2.4 Operating Environment

The system functions within a contemporary, web-based environment enhanced by artificial intelligence, supporting intelligent image analysis, personalized recommendations, and interactive user experiences. It is built to operate seamlessly across various devices, including desktops, laptops, tablets, and smartphones.

The front-end interface leverages modern web development tools such as Next.js and responsive design frameworks to ensure consistent performance and usability across diverse screen sizes and device types. Access is available through widely used web browsers like Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

On the back end, technologies including Python and FastAPI manage core operations such as AI-driven processing, recommendation logic, API interactions, and integration with other system components. These services are deployed on cloud infrastructure or local servers equipped to handle machine learning workloads and real-time data processing.

The AI-powered body measurement component relies on a custom-trained YOLO pose estimation model, with the Python environment managed via Anaconda. Image analysis and computer vision tasks are executed using OpenCV and NumPy. High-performance computing resources—preferably GPU-accelerated or cloud-based—are necessary to optimize model training and inference speeds. Model development and training are carried out using Google Colab, while annotation tasks are performed with CVAT.

The natural language processing module for recommendations utilizes sentence-transformer models, semantic similarity techniques, and machine learning workflows. This requires sufficient memory, computational capacity, and storage to efficiently process fashion-related datasets and generate accurate suggestions.

The CVD-focused outfit matching component operates in an AI-enabled environment compatible with both web and mobile platforms, using Python, OpenCV, and YOLO for detecting and analyzing clothing colors. It depends on internet connectivity, image upload functionality, and cloud-based computation to identify types of color vision deficiency and assess suitable color combinations. The interface incorporates accessibility features to improve usability for individuals with color blindness.

User data, recommendation records, clothing databases, and system settings are securely stored using MongoDB. Data protection is ensured through secure authentication protocols and encryption of stored information.

A stable internet connection is essential for cloud communication, accessing AI models, synchronizing databases, and delivering online recommendation services. Moderate bandwidth is adequate for efficient image uploads and timely generation of recommendations.

Overall, the operating environment is structured to support scalability, reliable performance, and real-time AI processing, while remaining compatible with a wide range of hardware and software setups.

2.5 Design and Implementation Constraints

The design and development of the proposed system are shaped by a range of technical, operational, and practical limitations.

A key challenge lies in the need for high-quality full-body images to accurately extract body measurements and predict sizes. Factors such as suboptimal lighting, improper posture, low image resolution, or partial body visibility can negatively impact the performance of the YOLO-based pose estimation method.

Equally important is the availability and reliability of training data. The accuracy of outfit suggestions, size predictions, and color assessments depends heavily on datasets that are diverse, well-annotated, and representative. To ensure models perform fairly and generalize well, they must be trained on data covering a broad spectrum of body types, genders, clothing styles, use cases, and color combinations.

Computational demands also present a significant hurdle. Training specialized models such as customized YOLO networks, NLP embedding systems, and image analysis tools requires substantial processing power, typically provided by GPUs or cloud-based infrastructure. Limited access to such resources can slow down training and hinder model refinement.

Real-time performance adds another layer of complexity. As the system delivers dynamic recommendations, it must efficiently handle image processing, natural language queries, and recommendation workflows at once. Maintaining fast response times becomes increasingly difficult as data volume and user activity grow.

Usability and accessibility are critical considerations. The interface must be intuitive and clearly presented for all users, including those with color vision deficiencies, ensuring equitable access and ease of use across varying visual capabilities.

System functionality also relies on consistent internet connectivity, given its dependence on cloud-hosted AI services, databases, and recommendation engines. Any disruption in network access could impair performance or limit availability.

Data privacy and security are paramount. Personal information and user-submitted images must be protected through secure storage, encryption, and proper authentication protocols to prevent unauthorized access or data breaches, while adhering to ethical data practices.

Cross-platform compatibility is essential as well. The system must operate smoothly across various devices, operating systems, and web browsers, delivering consistent responsiveness and functionality regardless of the user's setup.

Lastly, due to the project's academic and research-oriented scope, initial versions may not include support for every global fashion brand, large-scale commercial deployment, or multilingual recommendation features.

2.6 User Documentation

The proposed system will offer thorough user documentation to support effective understanding and operation of the platform, catering to both technical and non-technical users to ensure accessibility and ease of use.

A comprehensive User Manual will outline the full system workflow, covering account registration, login processes, image upload procedures, size analysis, recommendation generation, and outfit matching features. It will include step-by-step guidance, screenshots, and clear explanations of each function to aid user comprehension.

To assist new users in getting started quickly, a Quick Start Guide will be available, summarizing core functionalities and enabling efficient initial interaction with the platform.

For the AI-driven body measurement component, documentation will detail best practices for capturing and uploading full-body images, including advice on image quality, proper posture, and optimal lighting conditions necessary for accurate size estimation.

Users will also receive guidance on how to input detailed clothing preferences such as occasion, color, style, and fashion needs to improve the accuracy of outfit recommendations generated by the system.

Specialized instructions will be provided for users with Color Vision Deficiency, explaining how to select their specific type of color blindness, interpret color analysis outcomes, and understand the system's outfit suggestions accordingly.

In addition to user-facing materials, technical documentation will be developed for developers and administrators, covering system architecture, API integration, database design, AI model settings, deployment steps, and maintenance protocols.

Support resources such as troubleshooting guides and a Frequently Asked Questions (FAQ) section will address common issues related to image uploads, login difficulties, recommendation delays, and navigation challenges.

Collectively, the documentation is intended to enhance user understanding, accessibility, usability, and long-term maintainability of the AI-powered fashion assistance platform.

2.7 Assumptions and Dependencies

The development of the proposed system relies on a set of assumptions and external dependencies essential for its correct operation and successful deployment. A key assumption is that users will submit clear, well-framed full-body images, enabling precise extraction of body measurements and accurate size suggestions. The system expects these images to clearly show the user's body structure, with minimal obstructions and appropriate posture.

It is also assumed that users will enter truthful personal information, including clothing preferences and color vision details, during registration and when receiving recommendations. Inaccurate or missing data may lead to less relevant suggestions and reduced personalization effectiveness. Access to internet-connected devices such as smartphones, tablets, or computers capable of uploading images and interacting with web-based AI tools is considered a prerequisite for system use.

The quality of recommendations is closely tied to the availability of current, well-organized fashion data. As such, the system depends on regularly updated clothing databases, size guides, occasion labels, and fashion-related metadata to generate reliable outputs.

The project relies on third-party technologies, including YOLO-based pose detection frameworks, OpenCV, natural language processing models, machine learning tools, and cloud platforms. Significant changes in compatibility or disruptions in these services could impact system performance.

Adequate computational resources such as GPU acceleration, cloud hosting, data storage, and a robust backend infrastructure are necessary to support AI processing and real-time recommendation delivery. Stable internet connectivity is expected throughout image upload, recommendation processing, and communication with cloud services. Network instability may temporarily degrade performance and user experience.

Data security and privacy are contingent on effective database administration, secure authentication protocols, and protected hosting environments to safeguard personal data and uploaded images.

Lastly, ongoing enhancements including dataset expansion, model retraining, and system updates are anticipated to gradually improve recommendation accuracy, accessibility features, and overall functionality.

3. External Interface Requirements

3.1. user interfaces

3.1.1 Login registration

Style Sense AI

Welcome Back!

Log in to continue your personalized style journey.



Login

Enter your details to access your account

☐ Remember Me [Forgot Password?](#)

Login

Don't have an account? [Register](#)

Style Sense AI

Welcome Back!

Log in to continue your personalized style journey.



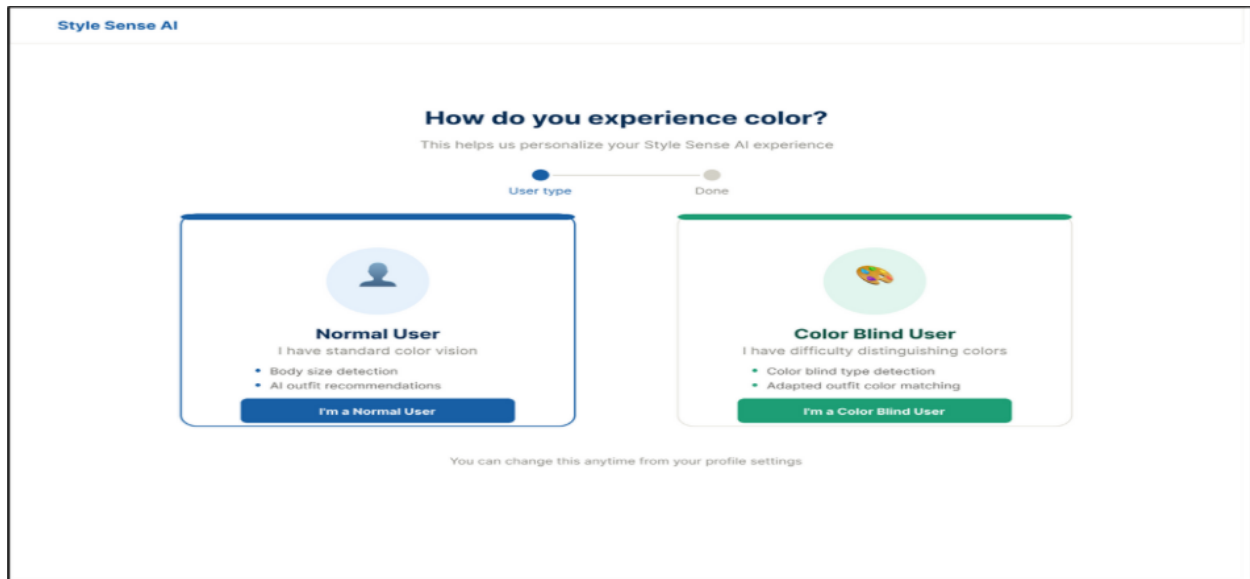
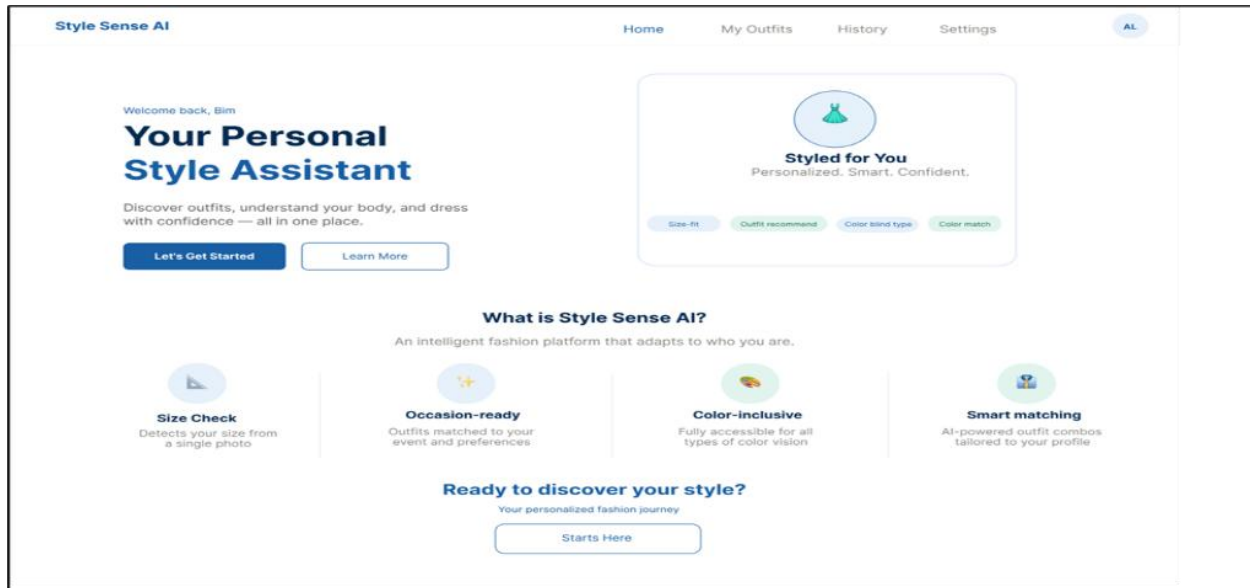
Register

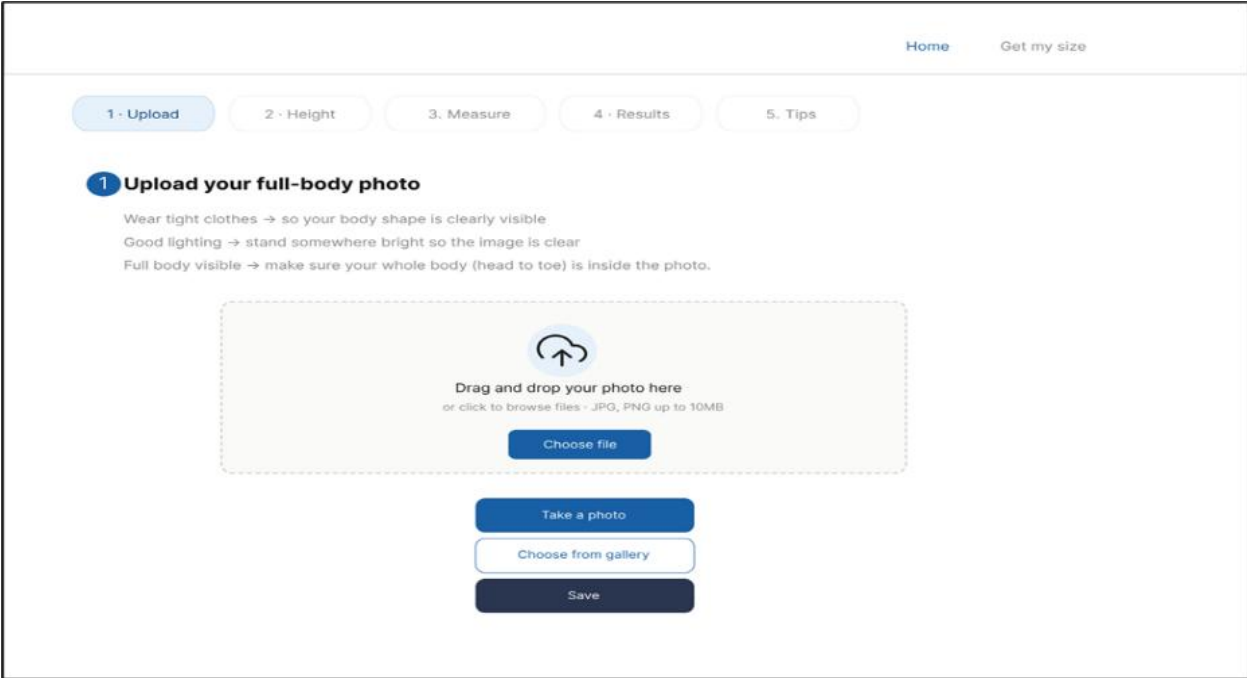
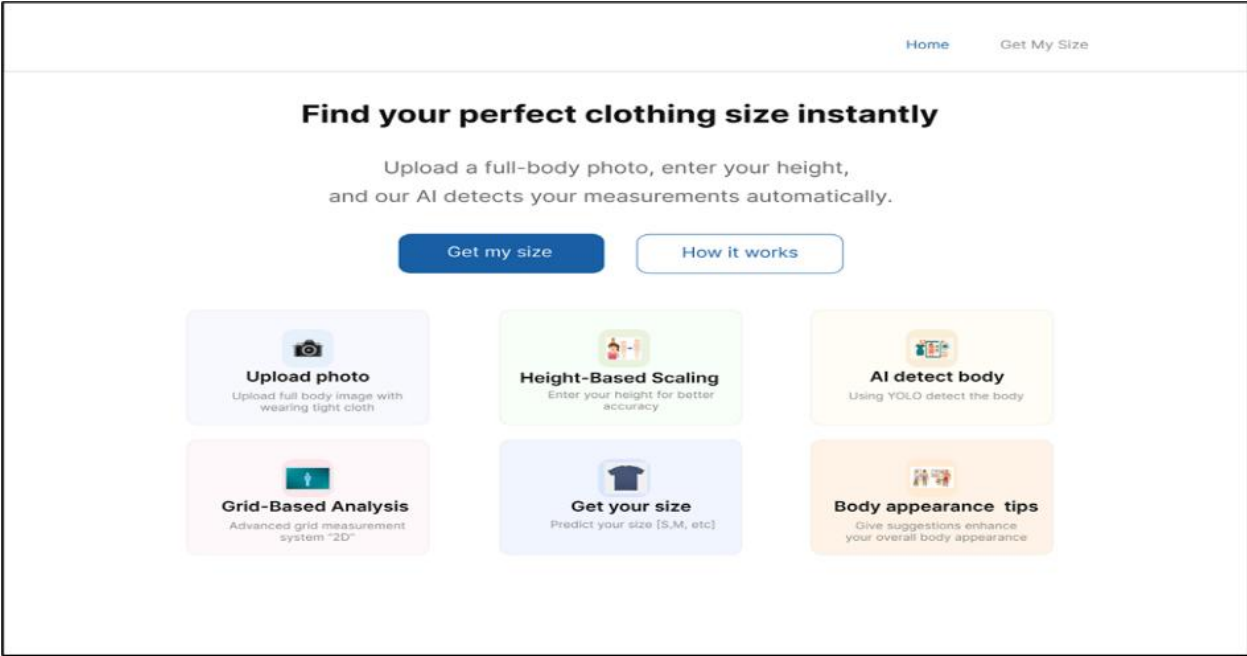
Fill your details to create your account

Register

Don't have an account? [Register](#)

3.1.2. AI-Based Virtual Intelligent Size Recommendation UI diagram





[Home](#)[Get my size](#)

1 · Upload ✓

2 · Height

3 · Measure

4 · Results

5 · Tips

2 Enter your real height

Used to create a pixel-to-real-world scale. This is the only manual input required.

Height (cm)

cm or ft in

Why height is needed

We use your height to understand your real body size.
Your photo gives us visual information, and your height helps us make it accurate.

Save

[Home](#)[Get my size](#)

1 · Upload ✓

2 · Height ✓

3 · Measure

4 · Result

5 · Tips

3 Person Body Measure

Automatically detect and measure your body dimensions for accurate size prediction



Body Measuring In Process

48% Completed

Extracting Person from Image

✓

Detecting Body & Calculating Grid Scale

✓

Measuring Your Shoulder, Hip & Torso Size



Check the result

[Home](#)[Get my size](#)

1 · Upload ✓

2 · Height ✓

3 · Measure ✓

4 · Result

5 · Tips

4

Your predicted Clothing Size

Matching your body dimensions to standard clothing size charts

M

Your size is Medium

Size chart match

Measurement	Your value	M range
Shoulder	43.5 cm	42~45 cm ✓
Hip width	36.2 cm	35~38 cm ✓
Height	172 cm	170~176 cm ✓
All 3 measurements agree → Medium		

How size was predicted

Shoulder width, hip width, and height fed into a trained size classifier. Majority vote across 3 measurements → Medium (all 3 agree).

Check my results

check my body tips

[Home](#)[Get my size](#)

1 · Upload ✓

2 · Height ✓

3 · Measure ✓

4 · Result ✓

5 · Tips

5

Body Appearance Tips

Based on your body measurements, here are personalized tips to improve your overall appearance.

Detected via photo analysis

BMI 22.4 — Normal Range

AGE24 years old

HEIGHT175 centimetres

BODY SIZE
M Medium fit

BMI RANGE
22.4 Normal

PERSONALISED BODY MAINTENANCE TIPS

☰

Healthy Weight Maintenance

Based on your height and current BMI

1

Ideal weight range for 175 cm is approx **65~75 kg**— you are in a healthy zone

2

Maintain **balanced diet with adequate protein** to support muscle tone and body definition

3

Your BMI of **22.4** is normal — focus on **maintaining** rather than losing weight

🕒

Daily Healthy Habits

Small changes for visible results

1

Drink at least **2.5 litres of water daily** — improves skin tone

2

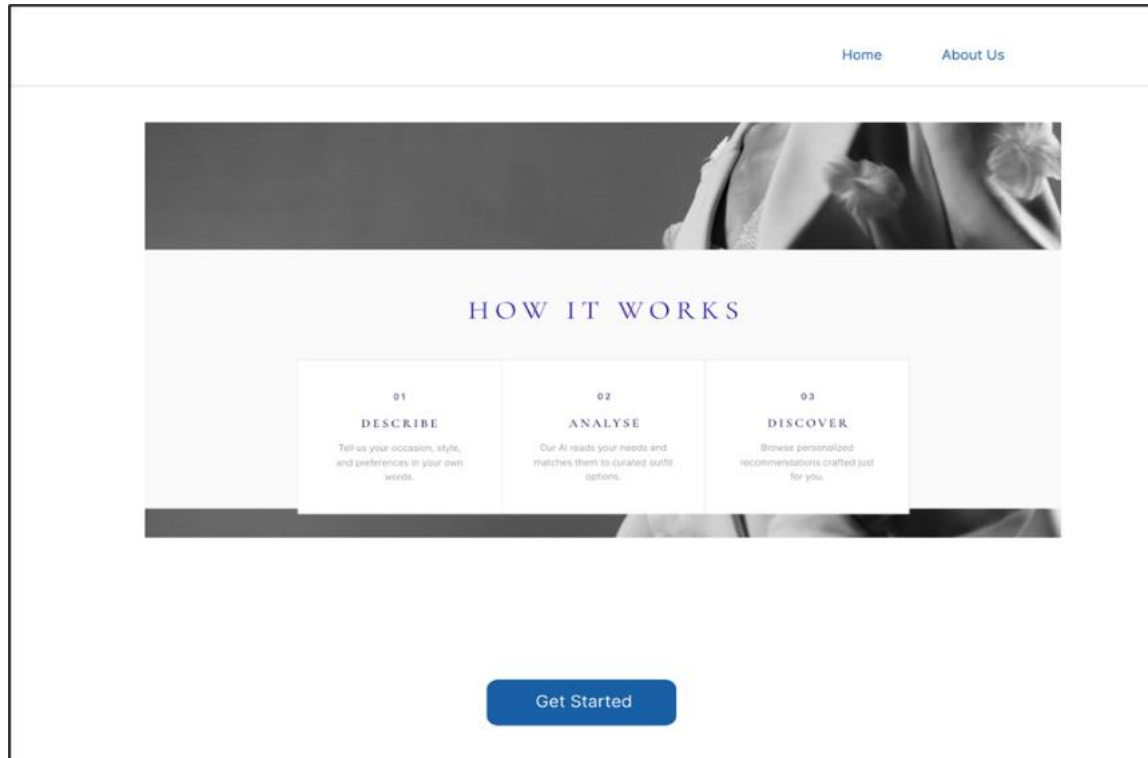
Get **7~8 hours of quality sleep** — body recovers and muscles improve

3

Walk at least **8,000 steps per day** — improves posture and circulation

Save

3.1.3 Intelligent Text-Driven Fashion Recommendation UI diagram



Tell Us What You Need

Describe your outfit requirements and we'll curate the perfect look.

DESCRIBE YOUR OUTFIT NEEDS

Describe your outfit needs...

E.g. I need a formal outfit for an office meeting in hot weather

52/500

READY ✓

TRY AN EXAMPLE

"A formal outfit for an office meeting in hot weather"

"Casual summer outfit for a beach day with friends"

"Party look for a rooftop event in the evening"

OCCASION

OPTIONAL FILTERS

CASUAL

FORMAL

PARTY

SPORTS

DATE NIGHT

BUSINESS

COLOR PREFERENCE

WHITE

BLACK

BLUE

RED

GREEN

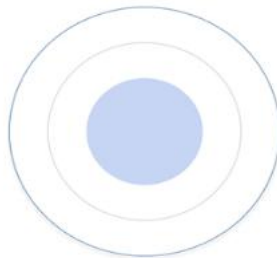
YELLOW

PINK

PURPLE

NEUTRAL

GET
RECOMMENDATIONS



Curating Your Looks

Curating the perfect wardrobe...

- ☒ Reading your outfit requirements...
- ☒ Analysing occasion & style context...
- ☒ Matching your preferences...
- ☐ Curating the perfect wardrobe...
- ☒ Finalising your looks...

Natural fabrics like cotton and linen breathe beautifully.

View

← BACK

AI ANALYSIS COMPLETE Your Curated Looks

"A formal outfit for an office meeting in hot weather"



Slim Linen Trousers ★ 4.5

Linen is one of the best fabrics for hot weather — naturally breathable and

\$79.99 [VIEW MORE](#)

FORMAL BREATHABLE



Casual Summer Co-ord Set ★ 4.8

Co-ord sets eliminate the guesswork — this one is specifically designed for hot, casual

\$95.99 [VIEW MORE](#)

CASUAL TRENDY



Light Cotton Linen Shirt ★ 4.8

Matches your need for a casual, breathable outfit in hot weather. The linen fabric wicks

\$45.99 [VIEW MORE](#)

CASUAL BREATHABLE

TRY ANOTHER SEARCH

← BACK TO RESULTS



✔ Ideal for hot weather — the rayon fabric keeps you cool while the flowy silhouette allows air movement around the body.

DRESS

Flowy Summer Sundress

★★★★☆ 4.7/5.0

\$59.99 ~~\$79.99~~ 23% OFF

MATERIAL

100% Rayon

OCCASION

Casual

A lightweight, floral-print sundress with adjustable straps. Made from 100% rayon for maximum comfort in the heat.

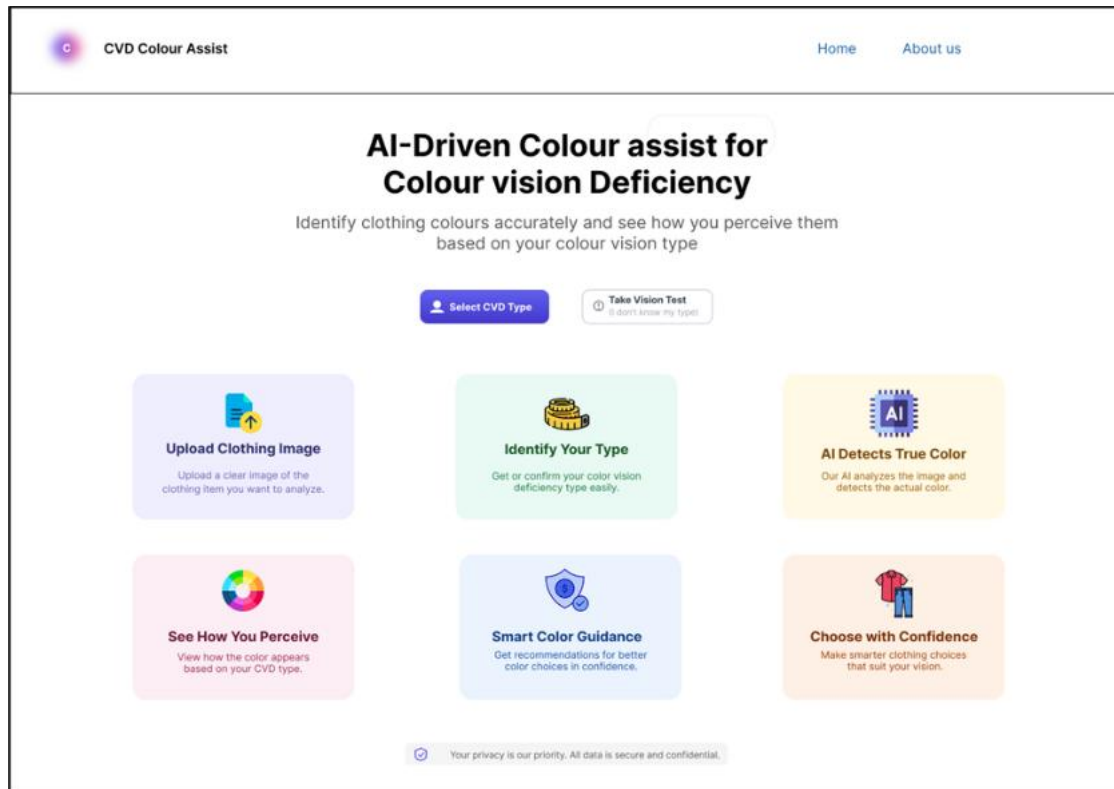
STYLE TIPS

Add a wide-brim hat for sun protection

Pair with flat sandals for a beach-ready look

Layer a light denim jacket for evenings

3.1.4 AI-Driven Color Assist for Color Vision Deficiency



C

CVD Colour Assist

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STEP 01 OF 05

Select Your Vision Type

Tell us how you see the world so we can help you see colors clearly

Red Deficiency



Protanopia

Difficulty seeing red wavelengths
Red ↔ Green confusion

looks like

Select Protanopia

✓ SELECTED

Green Deficiency



Deuteranopia

Difficulty seeing green wavelengths
Blue ↔ Purple confusion

looks like

✓ Selected

Blue Deficiency



Tritanopia

Difficulty seeing blue wavelengths
Blue ↔ Yellow confusion

looks like

Select Tritanopia

Not sure which type you have?

[Take the Ishihara Test →](#)

Continue with Deuteranopia →

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STEP 03 OF 05 — UPLOAD IMAGE

Upload Your Clothing

Upload a clear photo of your clothing item for AI color analysis



Drag & Drop your image here
or click to browse files

Supports JPG, PNG, WEBP - Max 10MB

Choose File

or

Take a Photo



1_shirt.jpg

2.4 MB · 1280 × 960 px · JPG

✓ Ready

Tip: Use good lighting for accurate color detection

X Remove

↺ Retake

 **Tips for best results**

- Use natural daylight or bright indoor lighting
- Lay the garment flat or hang it
- Avoid harsh shadows or reflections

Analyze Color with AI →

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STEP 04 OF 05 — AI DETECTION

Color Detection Complete

Our AI has analyzed your clothing image and identified the actual color

UPLOADED IMAGE



→



Green

#20B757

See How You Perceive This Color →

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STEP 05 OF 05 — YOUR RESULTS

Color Perception Result

Here's how the actual color compares to what you perceive with Deuteranopia

Vision type: Deuteranopia



Green

#20B757

VS

DETECTED COLOR



Yellow

#E9F241

Analyze Another Item

Export Results as PDF

 CVD Colour Assist

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Recent Scans

Green striped shirt
2 visible patterns. High contrast.

Beige checked trousers
Low CVD contrast.Enhanced.

Coral polka Dress
Pattern detected. Solid colour.



Drag & Drop your image here
or click to browse files
Supports JPG, PNG, WEBP / Max 10MB

Choose File

or

Take a Photo

Analyze pattern →

☒ Your privacy is our priority. All data is secure and confidential.

 CVD Colour Assist

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Pattern Analysis

Green Stripped shirt - Deuteranopia

Before vs After Enhancement

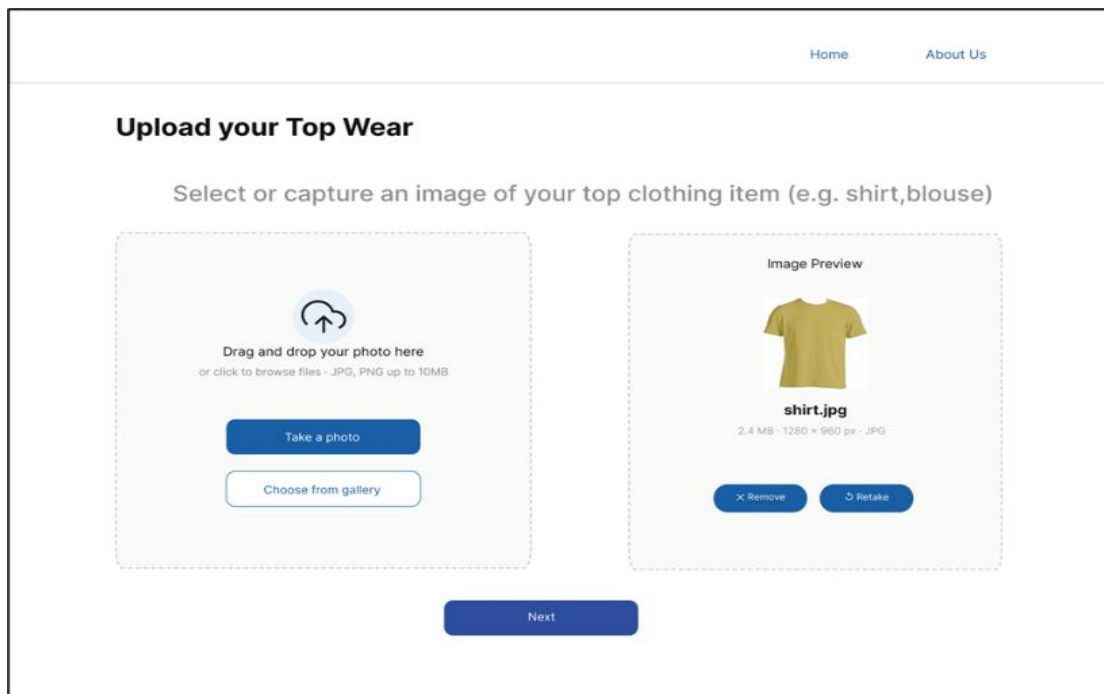
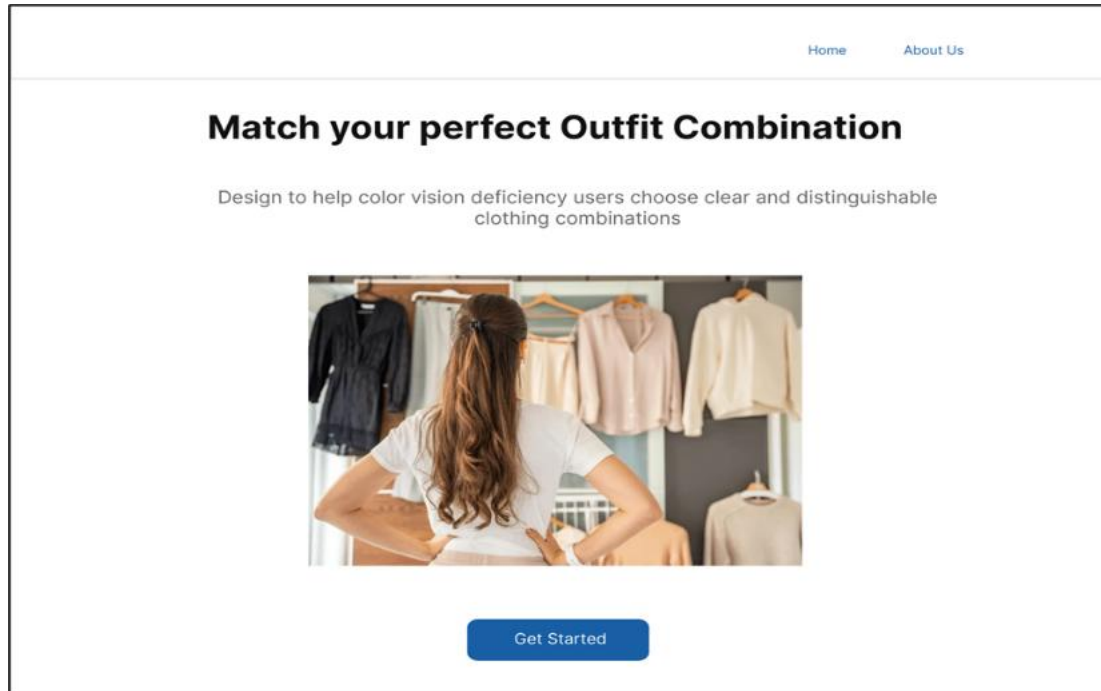
Original Normal trichromat vision	CVD perception Deuteranopia simulation	Enhanced Hue + luminance optimised
		
Stripes — easy Multi-color Pleat — medium	Red/green — lost Multi-color — hard Blues — retained	All patterns — clear High-contrast CVD-safe hues



High contrast enhanced.
Pattern clearly identifiable

☐ Your privacy is our priority. All data is secure and confidential.

3.1.5 Adaptive Outfit Matching for Color Vision Deficiency Users UI diagram



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Upload your Bottom Wear

Select or capture an image of your bottom clothing item (e.g. shorts, trousers)



Drag and drop your photo here
or click to browse files - JPG, PNG up to 10MB

Take a photo

Choose from gallery

Image Preview

denim.jpg
2.4 MB · 1280 × 960 px · JPG

X Remove

↻ Retake

Check matching results

[Home](#)[About Us](#)

Match Results

Top Wear



Bottom Wear



Match result Good match

Delta E score: **57.43** — These colours pair well together — good contrast.

VISIBILITY UNDER YOUR VISION TYPE

- Top (Orange) — hard to distinguish under Deuteranopia.
- Bottom (Blue) — distinguishable under Deuteranopia.

Exit

Match Results

Top Wear



Black

#E934671

Bottom Wear



Light Blue

#20B757

Match result

Poor match

Delta E score: **17.11** — These colours are too similar or clash — consider changing one.

VISIBILITY UNDER YOUR VISION TYPE

- Top (Black) — distinguishable under Deuteranopia.
- Bottom (Blue) — distinguishable under Deuteranopia.

Check Alternatives

Exit

Check Alternatives

Here are alternative colour combinations that may offer better visibility under Deuteranopia.

Your Current Match



Poor match

Delta E score: **17.11**

These colours are too similar or clash — consider changing one.

Better Alternatives

Best match



White + Dark Blue

#FFFFFF #0D1B2A



Black + Light Blue

#000000 #89CFF0



Navy + Beige

#1B2A4E #F5F5DC

3.2 Hardware Interfaces

- Laptop with minimum 8GB RAM (recommended for AI model processing)
- GPU-enabled system (improves performance for deep learning tasks)
- Local storage / SSD for dataset storage, model files, and system backups
- Camera or mobile device support for capturing full-body and clothing images

3.3 Software Interfaces

- Database Management System (MongoDB) for storing user data, images, and recommendations
- Authentication service for secure login and user management
- AI/ML Model APIs (YOLO pose estimation, CNN models for color and pattern detection, NLP recommendation model)
- Image processing libraries (ex: OpenCV) for enhancement and analysis

3.4 Communications Interfaces

- HTTPS protocol for secure communication between client and server
- RESTful API for interaction between frontend and backend services
- JSON format for structured data exchange between system components

4. System Features

4.1 AI Based Virtual Intelligent Size Recommendation System

4.1.1 Description and Priority

Feature ID	Feature Name	Description	Priority
SF1	User Image Upload	The system allows users to upload a full-body image through the user interface for body measurement analysis.	High
SF2	Human Key Point Detection	The system detects 17 human body key points automatically using the custom-trained YOLO pose estimation model.	High
SF3	Body Measurement Extraction	The system calculates pixel-based measurements such as shoulder width, hip width, and estimated body height from detected key points.	High
SF4	Real-World Measurement Estimation	The system converts pixel distances into estimated real-world body measurements using body ratios and scaling methods.	High
SF5	Clothing Size Recommendation	The system compares extracted measurements with predefined clothing size charts and recommends suitable clothing sizes.	High
SF6	Body Type Analysis	The system analyzes body proportions and body structure to identify body appearance characteristics.	Medium
SF7	Personalized Appearance Tips	The system provides body appearance and fashion-related suggestions according to the detected body proportions.	Medium
SF8	Dataset and Model Management	The administrator can manage datasets, trained models, and system records for improving model performance.	Medium

SF9	Result Visualization	The system displays the uploaded image with detected key points and measurement outputs visually.	Medium
SF10	Secure Data Handling	The system securely stores uploaded images and measurement data while protecting user privacy.	High

4.1.2 Stimulus/Response Sequences

Stimulus	System Response
User enters height and uploads a full-body image	The system validates the entered height and uploaded image before processing.
User clicks the “Predict” button	The system sends the image to the YOLO pose estimation model for key point detection.
YOLO model detects body key points	The system identifies 17 human body key points and marks them on the image.
Key points are successfully detected	The system calculates pixel-based measurements such as shoulder width, hip width, and body height.
User height is available	The system uses the user’s real height to convert pixel measurements into estimated real-world body measurements.
Measurement conversion is completed	The system compares extracted measurements with clothing size charts.
Size comparison is completed	The system recommends the most suitable clothing size for the user.
Body ratio analysis is completed	The system generates personalized body appearance and fashion tips.
Invalid image or missing height input	The system displays an error message requesting valid height details and a clearer full-body image.

4.1.3 Functional Requirements

Requirement ID	Functional Requirement Description
FR1	The system shall allow users to enter their height and upload full-body images.
FR2	The system shall validate uploaded images and user height before processing.
FR3	The system shall detect 17 human body key points using the custom-trained YOLO pose estimation model.
FR4	The system shall calculate pixel-based body measurements such as shoulder width, hip width, and body height.
FR5	The system shall convert pixel measurements into estimated real-world measurements using the user's height.
FR6	he system shall compare extracted measurements with clothing size charts.
FR7	The system shall recommend suitable clothing sizes for users.
FR8	The system shall display detected key points, body measurements, and size recommendation results through the user interface.

4.2 Intelligent Text-Driven Fashion Recommendation

4.2.1 Description and priority

Feature ID	Feature Name	Description	Priority
SF1	Free-Text Fashion Query Input	Users can enter clothing requirements in natural language such as style, occasion, colors, and preferences.	High
SF2	NLP-Based Semantic Analysis	The system processes user text using NLP and sentence embeddings to understand the meaning of the query beyond keywords.	High
SF3	Fashion Recommendation Engine	Matches user requirements with relevant clothing items from the dataset using semantic similarity.	High
SF4	Size Prediction and Filtering	Integrates body measurement and size prediction to ensure recommended clothes fit the user correctly.	High
SF5	Occasion-Aware Ranking	Prioritizes clothing items based on events such as weddings, office wear, casual outings, or festivals.	High
SF4	Gender-Inclusive Recommendations	Supports both ladies' and gents' fashion recommendations.	Medium
SF5	Product Dataset Management	Maintains and updates clothing datasets including descriptions, sizes, colors, and categories.	Medium
SF6	Recommendation Explanation Module	Explains why a specific item was recommended to improve user trust and transparency.	Medium

4.2.2 Stimulus/Response Sequences

Scenario	Stimulus (User Action / System Event)	System Response
Fashion Preference Search	User enters a clothing request in natural language	The NLP module analyzes the text and extracts style, occasion, and preference information.
Occasion-Based Recommendation	User selects an occasion such as wedding or office	The recommendation engine boosts relevant clothing items matching the occasion.
Recommendation Generation	User clicks the “Recommend” button	The system retrieves and displays the most relevant clothing recommendations.
Error Handling	User enters invalid or incomplete input	The system displays an error message and requests valid information.

4.2.3 Functional Requirements

ID	Requirement
FUN-1	The system shall accept unstructured natural language text as user input for preferences, occasion, and style.
FUN-2	The system shall use Sentence-Transformers or equivalent embedding models to process user text semantically.
FUN-3	The system shall integrate size prediction or filtering as a core step before or during ranking.
FUN-4	The system shall perform semantic matching between user queries and product descriptions.
FUN-5	The system shall support both male and female users with appropriate recommendations.

FUN-6	The system shall provide explainable output (e.g., “recommended because it matches your occasion and style preferences”).
-------	---

4.3. AI-Driven Color Assist for Color Vision Deficiency (CVD)

4.3.1 Description and Priority

Feature ID	Feature Name	Description	Priority
SF1	Clothing Image Upload	Users can upload clothing images to analyze colors and patterns.	High
SF2	AI-Based Color Identification	The system detects the dominant colors in the clothing image using a CNN-based color classification model.	High
SF3	CVD Simulation	Simulates how the clothing appears to users with Protanopia, Deuteranopia, and Tritanopia.	High
SF4	Pattern Enhancement	Applies image processing techniques such as contrast enhancement and overlays to make clothing patterns more distinguishable.	High
SF5	Color Description Generation	Provides easy-to-understand color labels such as “dark blue with white stripes.”	High
SF6	Accessibility Feedback	Displays recommendations and visual explanations to help users better understand the garment.	Medium
SF7	Support for Multiple CVD Types	Allows users to select their specific type of color vision deficiency.	Medium
SF8	Image Dataset Management	Maintains and updates the clothing image dataset used for model training and testing.	Medium

4.3.2 Stimulus/Response Sequences

Scenario	Stimulus (User Action / System Event)	System Response
Clothing Image Analysis	User uploads a clothing image	The system detects dominant colors and patterns in the garment.
CVD Simulation	User selects a color vision deficiency type	The system generates a simulated image showing how the garment appears to that user.
Pattern Enhancement	User requests enhanced visualization	The system improves contrast and adds overlays to highlight important patterns.
Color Description	Image analysis is completed	The system displays descriptive labels for detected colors and patterns.
Error Handling	User uploads an invalid or unsupported image	The system displays an error message and asks for a valid image file.

4.3.3 Functional Requirements

ID	Requirement
FUN-1	The system shall allow users to upload clothing images in standard image formats such as JPG and PNG.
FUN-2	The system shall identify clothing colors using a trained deep learning model.
FUN-3	The system shall simulate clothing appearance for Protanopia, Deuteranopia, and Tritanopia.
FUN-4	The system shall enhance clothing patterns using image processing techniques such as contrast enhancement and overlays.
FUN-5	The system shall generate textual descriptions of detected colors and patterns.
FUN-6	The system shall allow users to choose their color vision deficiency type.

FUN-7	The system shall display both the original and enhanced images for comparison.
FUN-8	The system shall notify the user when an uploaded image cannot be processed.

4.4. Adaptive outfit matching for color vision deficiency users

4.4.1 Description and Priority

Feature ID	Feature Name	Description	Priority
SF1	Outfit Image Input	Users can upload or select top and bottom clothing images for compatibility checking.	High
SF2	Color Extraction for Outfit Pieces	The system extracts dominant colors from both top and bottom clothing using AI-based color detection.	High
SF3	CVD-Based Outfit Simulation	Simulates how the combined outfit appears under Protanopia, Deuteranopia, and Tritanopia conditions.	High
SF4	Color Compatibility Matching	Evaluates whether the selected top and bottom colors are visually compatible	High
SF5	Contrast & Visibility Scoring	Provides a score indicating how clearly the outfit elements can be distinguished under different CVD types.	Medium
SF6	Style & Pattern Awareness	Considers patterns (stripes, prints, gradients) in addition to solid colors for matching decisions.	Medium

4.4.2 Stimulus/Response Sequences

Scenario	Stimulus (User Action / System Event)	System Response
Outfit Upload	User uploads or selects top and bottom clothing images	The system extracts dominant colors and pattern features from both items.
Compatibility Check	User clicks “Check Outfit Match”	The system analyzes color harmony and evaluates compatibility for CVD users.

CVD Simulation View	User selects a CVD type	The system displays how the full outfit appears under the selected vision deficiency.
Score Display	Analysis is completed	The system shows a compatibility score with explanation (e.g., low contrast, high mismatch risk).
Invalid Input Handling	User uploads unsupported or unclear images	The system shows an error message and requests valid clothing images.

4.4.3 Functional Requirements

ID	Requirement
FUN-1	The system shall allow users to upload or select both top and bottom clothing images.
FUN-2	The system shall extract dominant colors and patterns from both clothing items using AI models.
FUN-3	The system shall evaluate color compatibility between top and bottom clothing for CVD users.
FUN-4	The system shall simulate outfit appearance under Protanopia, Deuteranopia, and Tritanopia conditions.
FUN-5	The system shall generate outfit compatibility scores based on color contrast and visibility.
FUN-6	The system shall consider both color and pattern when analyzing outfit matching suitability.
FUN-7	The system shall notify users when uploaded outfit images cannot be processed or analyzed.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should guarantee that user-uploaded images and personal data are handled safely at all times during system operation. The uploaded images of a person's body and clothes should only be utilized for the purposes of sizing, outfit recommendation, and color analysis. The system should have measures in place to avoid any uploads that are inappropriate or harmful in nature. There should also be guidelines provided to the users on how to take pictures of their whole bodies to avoid incorrect measurement and recommendations. Moreover, the recommendations provided by the system should be uplifting and respectful and should not lead to any form of discrimination or offensive comments related to the body of the user.

5.2 Safety Requirements

The system should guarantee that user-uploaded images and personal data are handled safely at all times during system operation. The uploaded images of a person's body and clothes should only be utilized for the purposes of sizing, outfit recommendation, and color analysis. The system should have measures in place to avoid any uploads that are inappropriate or harmful in nature. There should also be guidelines provided to the users on how to take pictures of their whole bodies to avoid incorrect measurement and recommendations. Moreover, the recommendations provided by the system should be uplifting and respectful and should not lead to any form of discrimination or offensive comments related to the body of the user.

5.3 Security Requirements

This will ensure that there is an appropriate authentication and authorization process in place to prevent any kind of intrusion into the system by securing user accounts and preventing any kind of leakage of the information that has been stored on the system. Passwords for each user account should be encrypted prior to storing them in the database. Information that includes uploaded images, personal information, and the recommendation history of users shall be safely

stored through the use of protected databases and secure cloud storage solutions. This system will utilize secure channels of communication like HTTPS for transmission of the data to and from the server.

5.4 Software Quality Attributes

The suggested system will adhere to high software quality metrics in order to guarantee that the application is reliable, usable, maintainable, and scalable. The system will deliver correct and consistent recommendations for sizing clothing, outfit suggestions, and assistance with color blindness under typical conditions. The user interface will be designed in such a way as to be intuitive and easy-to-use regardless of the experience level of a user. At the same time, the suggested platform will possess the capabilities to support accessibility options that are appropriate for the case of Color Vision Deficiency. Additionally, the system architecture will be modular and maintainable, providing room for upgrades.

5.5 Business Rules

The system shall require users to create an account and log in before accessing personalized recommendation services and AI analysis features. Users must upload a valid full-body image to obtain accurate clothing size predictions and body appearance analysis results. The system shall generate outfit recommendations according to user preferences, predicted body size, occasion requirements, and selected user category. Users identified as normal users shall receive standard outfit recommendations, while users identified as color-blind users shall receive CVD-aware outfit matching and color accessibility assistance. The system shall not allow unauthorized access to recommendation datasets, AI models, or user information. Additionally, all recommendation processes shall follow ethical AI principles by providing fair, inclusive, and non-discriminatory fashion suggestions for all users regardless of body type, gender, or visual perception condition.

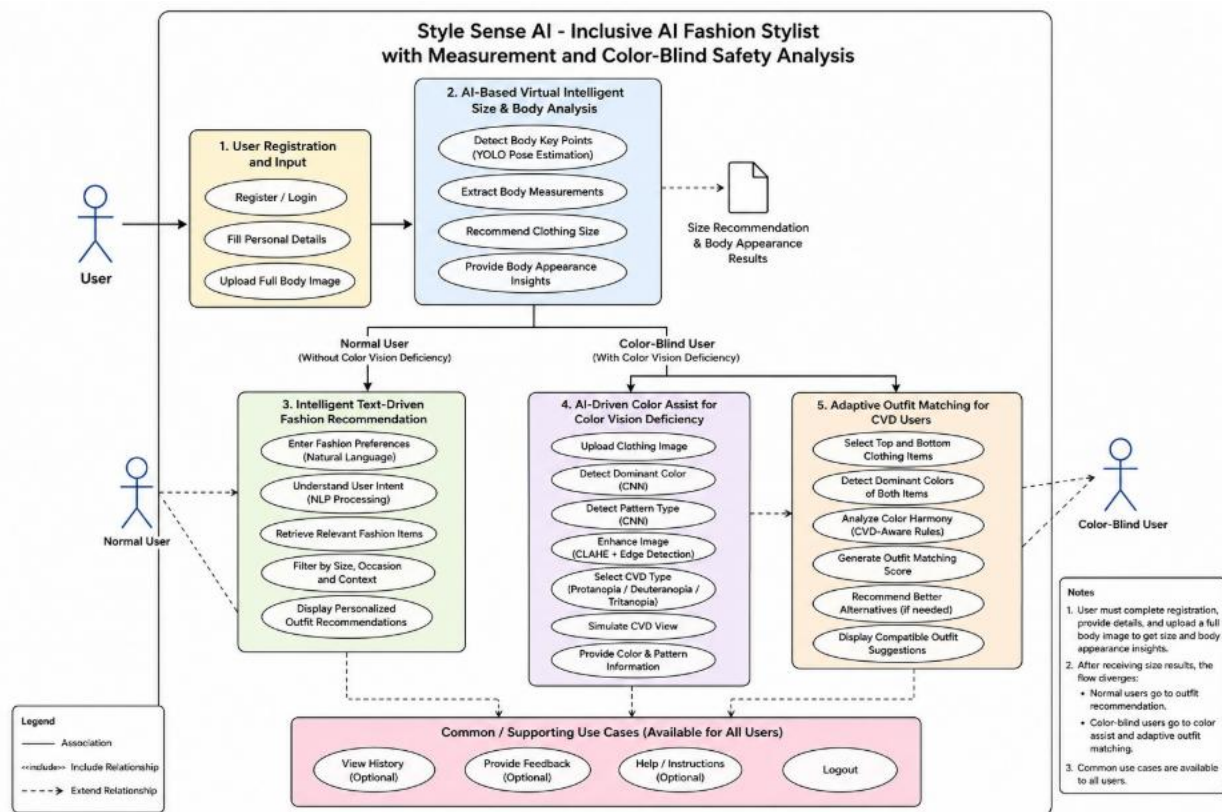
6. Appendix A: Glossary

Term	Definition
ML	Machine Learning (ML) is a branch of Artificial Intelligence that enables systems to learn patterns from data and improve performance without being explicitly programmed.
NLP	Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and process human language (text or speech) to extract meaning and generate useful outputs such as recommendations or responses.
YOLO	You Only Look Once (YOLO) is a real-time object detection algorithm used to detect objects and key body points in images.
CNN	Convolutional Neural Network (CNN) is a deep learning model used for image classification, feature extraction, color detection, and pattern recognition.
CVD	Color Vision Deficiency (CVD) is a condition where a person has difficulty distinguishing certain colors.
CLAHE	Contrast Limited Adaptive Histogram Equalization (CLAHE) is an image processing technique used to improve image contrast.
AI	Artificial Intelligence (AI) is the simulation of human intelligence in machines to perform tasks such as decision-making, learning, and problem-solving.

7. Appendix B: Analysis Models

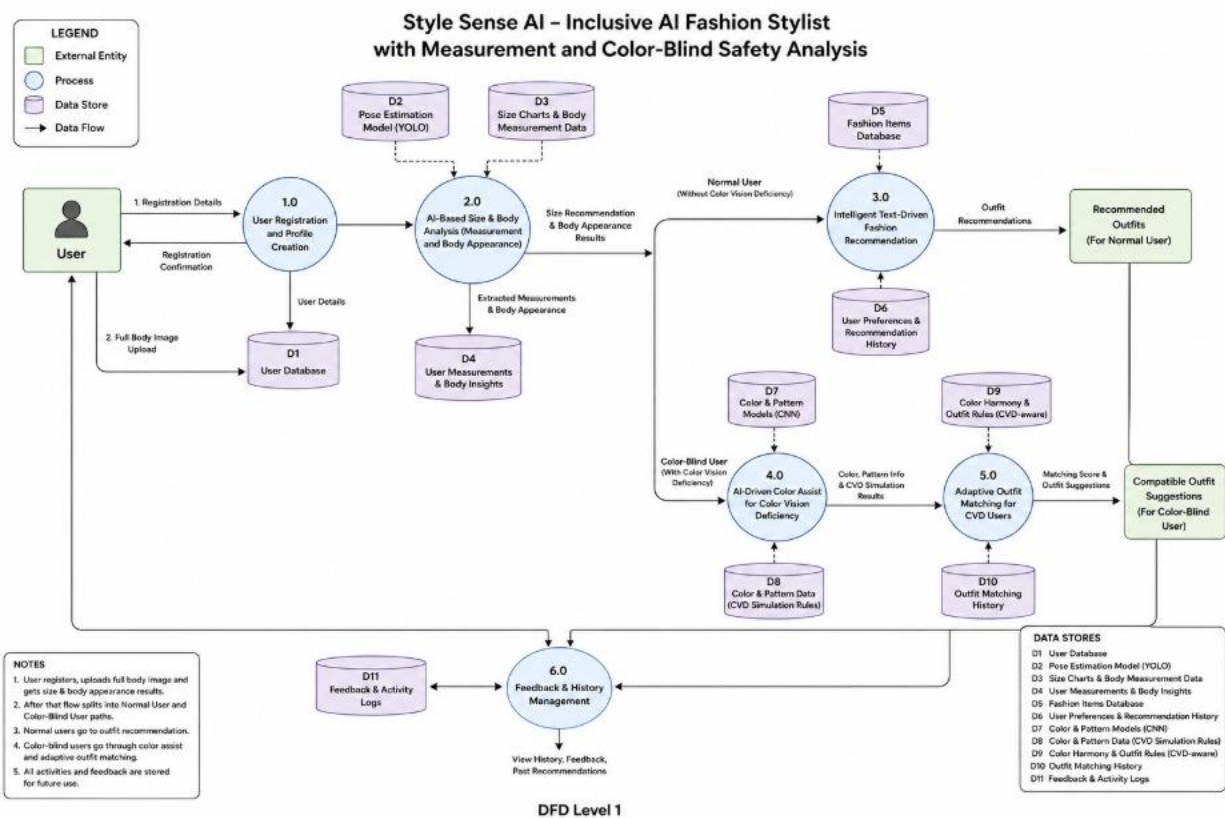
1. use case diagram

The Use Case Diagram illustrates how users interact with the Style Sense AI system. First, users register, enter personal details, and upload a full-body image to receive clothing size recommendations and body appearance insights through the AI-based body analysis module. After the size prediction stage, the system separates users into two categories: normal users and color-blind users. Normal users continue to the intelligent text-driven fashion recommendation module to receive personalized outfit suggestions based on their preferences and occasions, while color-blind users access the AI-driven color assist and adaptive outfit matching modules to receive color-safe and visually compatible outfit recommendations according to their specific CVD type.



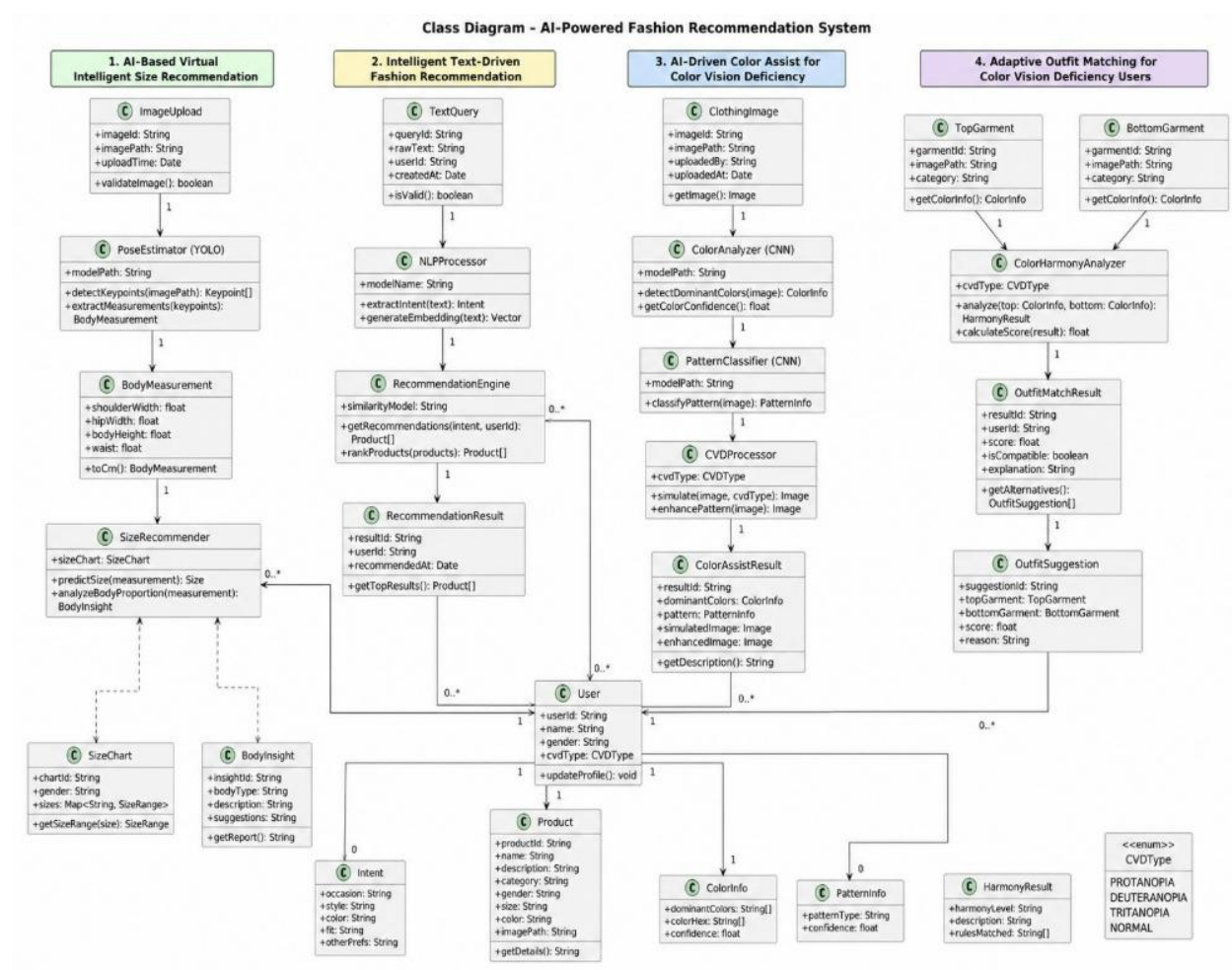
2.Data flow diagram

The DFD Level 1 diagram shows how data flows through the Style Sense AI system. Users provide registration details, upload images, and enter fashion preferences, which are processed by different AI modules. The body analysis module extracts measurements and predicts clothing sizes using YOLO pose estimation. After this, normal users receive outfit recommendations through the NLP-based fashion recommendation module, while color-blind users receive color analysis, CVD simulation, and adaptive outfit matching through specialized AI modules. All user information, measurements, recommendations, and feedback are stored in databases to improve personalization and system performance.



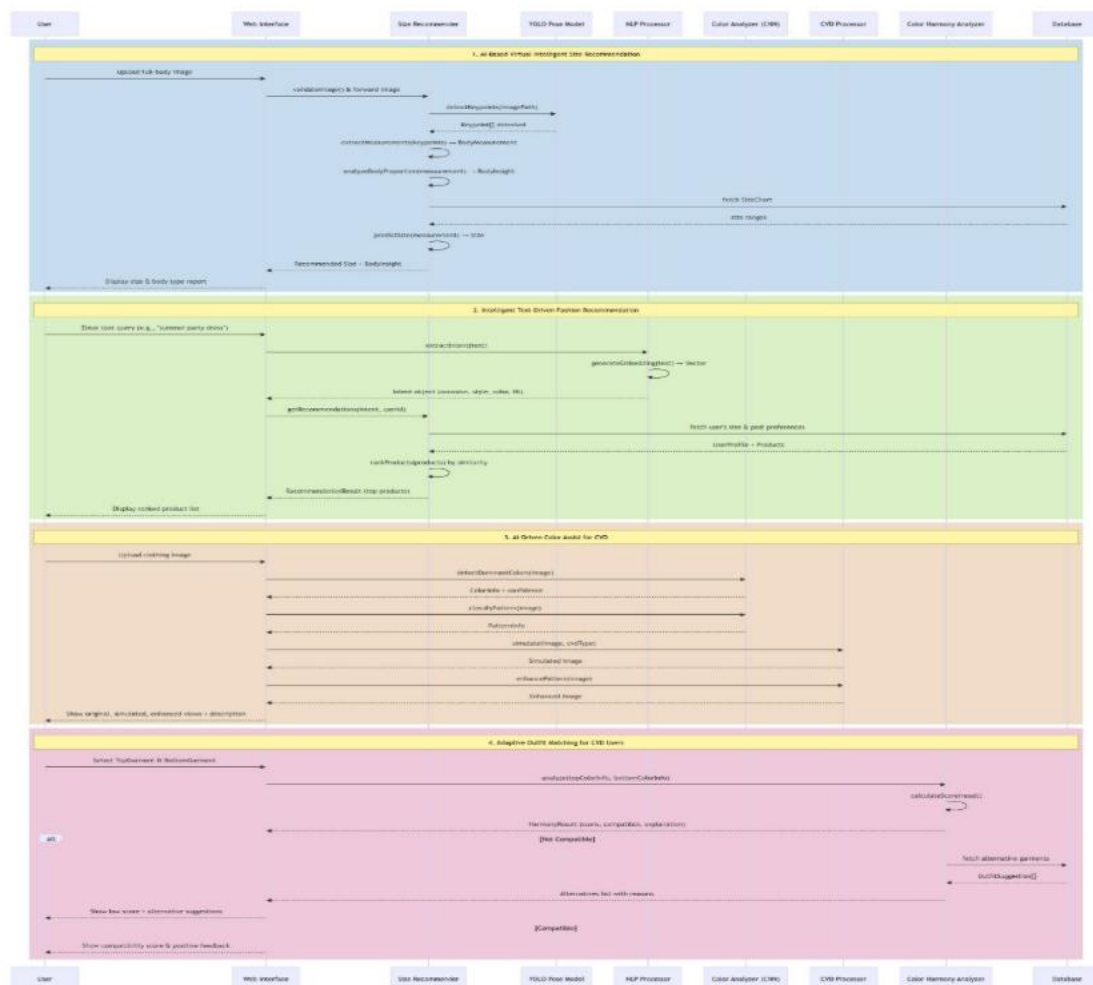
3. Class Diagram

Users interact with the system by uploading images, entering fashion preferences, and selecting their CVD type. The Size Recommendation Module predicts the most suitable clothing size from body measurements. The NLP Recommendation Module analyzes text input and retrieves personalized clothing suggestions. The Color Assist Module identifies colors and patterns and generates CVD-specific visual outputs. The Adaptive Outfit Matching Module evaluates outfit compatibility and recommends alternative combinations when needed. All modules share user, product, and recommendation data to provide an integrated and accessible fashion recommendation experience.



4. Sequence diagram

This sequence diagram shows how users interact with an AI-powered fashion recommendation system by uploading body images, clothing images, entering fashion preferences, and selecting their CVD type. The application server processes the request by coordinating modules such as size recommendation, NLP-based preference analysis, color assist, and outfit matching. The system uses a YOLO-based model for size prediction, NLP for understanding user preferences, and image processing for detecting colors and patterns with CVD support. Finally, the outfit matching module evaluates compatibility using color harmony rules and generates a matching score with alternative suggestions, which are displayed to the user through the web interface.



5. Activity diagram

